

Chapter 8

The Relational Algebra

1

Overview

A database is a set of relations and we can **operate** on **input** relations to form **new** relations.

The basic set of operators is the “closed” **relational algebra**

A sequence of relational algebra operations forms a **relational algebra expression**

Similar to **integer algebra**:

$$4+5 = 9 \quad 7-9 = -2 \quad 2*3 = 6 \quad ((4+5)*(2-3))/6$$

2

Relational Algebra

Consists of:

- Unary Operations
- Set Theoretic Operations
- Binary Operations

3

Operators

Unary

SELECT: σ
PROJECT: π
RENAME: ρ

Set Theoretic

UNION: $\cup$
INTERSECTION: $\cap$
DIFFERENCE: $-$
CARTESIAN PRODUCT: $\times$

Binary

JOIN: $\bowtie$
DIVISION: $\div$

4

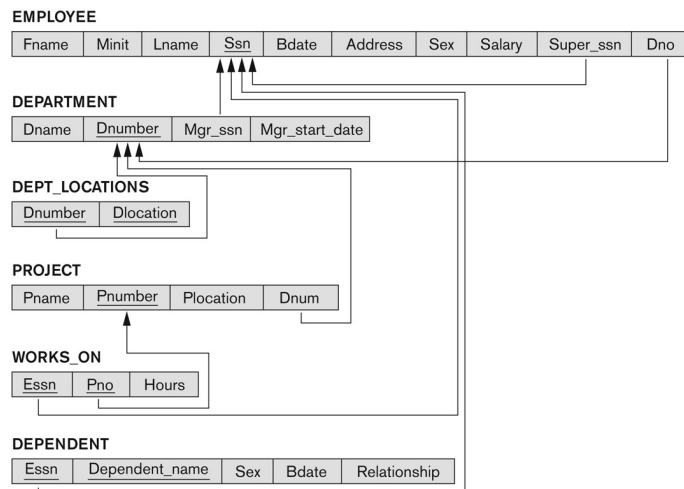
Functions

Aggregate

SUM
COUNT
AVG
MIN
MAX

5

COMPANY Database



6

The Database State

EMPLOYEE									
Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1982-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT			
Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

DEPT_LOCATIONS	
Dnumber	Dlocation
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

WORKS_ON		
Easn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

PROJECT			
Pname	Pnumber	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

DEPENDENT				
Easn	Dependent_name	Sex	Bdate	Relationship
333445555	Alice	F	1986-04-05	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

7

The SELECT Operator $\sigma_{cond}(R)$

The SELECT operator σ selects a **subset** of the tuples from a relation based on a **selection condition**.

- The selection condition is a **filter**
- Keeps only those tuples that satisfy the condition
- The result is a new relation with an identical schema

Select the EMPLOYEE tuples whose department number is 4:

$\sigma_{Dno=4}(\text{EMPLOYEE})$

Select the EMPLOYEE tuples whose salary is greater than \$30,000:

$\sigma_{Salary>30,000}(\text{EMPLOYEE})$

8

SELECT Properties

Commutativity:

$$\sigma_{cond1}(\sigma_{cond2}(R)) = \sigma_{cond2}(\sigma_{cond1}(R))$$

Conjunctivity:

$$\sigma_{cond1}(\sigma_{cond2}(R)) = \sigma_{cond1 \text{ AND } cond2}(R)$$

The number of tuples in $\sigma_{cond}(R)$ is $\leq$ the number of tuples in R .

9

The PROJECT Operator $\pi_{attr}(R)$

PROJECT keeps certain attributes of a relation and discards the others. *Duplicate tuples are removed.*

List each employee's first name, last name and salary:

$$\pi_{Fname,Lname,Salary}(EMPLOYEE)$$

10

PROJECT Properties

- The number of tuples in $\pi_{attr}(R)$ is $\leq$ the number of tuples in R .
- If $attr$ includes a **key** of R , then the number of tuples in $\pi_{attr}(R)$ is **equal** to the number of tuples in R
- If $attr$ contains an attribute not in $schema(R)$, then the result is undefined.

11

PROJECT Properties

PROJECT is commutative if all projections are well-defined:

$$\pi_{\langle list1 \rangle}(\pi_{\langle list2 \rangle}(R)) = \pi_{\langle list2 \rangle}(\pi_{\langle list1 \rangle}(R)) = \pi_{\langle list1 \cap list2 \rangle}(R)$$

If $\langle list1 \rangle \subseteq \langle list2 \rangle$ then

$$\pi_{\langle list1 \rangle}(\pi_{\langle list2 \rangle}(R)) = \pi_{\langle list1 \rangle}(R)$$

12

Examples

$\sigma_{(Dno=5 \text{ AND } Salary > 2500) \text{ OR } (Dno=5 \text{ AND } Salary > 3000)}(EMPLOYEE)$

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5

$\pi_{Lname, Fname, Salary}(EMPLOYEE)$

Lname	Fname	Salary
Smith	John	30000
Wong	Franklin	40000
Zelaya	Alicia	25000
Wallace	Jennifer	43000
Narayan	Ramesh	38000
English	Joyce	25000
Jabbar	Ahmad	25000
Borg	James	55000

$\pi_{Sex, Salary}(EMPLOYEE)$

Sex	Salary
M	30000
M	40000
F	25000
F	43000
M	38000
M	25000
M	55000

13

Single Expression vs. Sequence of Relational Operations

Retrieve the first name, last name, and salary of all employees who work in department #5:

$\pi_{Fname, Lname, Salary}(\sigma_{Dno=5}(EMPLOYEE))$

or

$DEP5EMP \leftarrow \sigma_{Dno=5}(EMPLOYEE)$

$RESULT \leftarrow \pi_{Fname, Lname, Salary}(DEP5EMP)$

14

The RENAME Operator $\rho_{S(attr)}(R)$

$\rho_{S(B_1, \dots, B_n)}(R)$ changes:

- the relation name to S , **and**
- the column (attribute) names to $B_1, \dots, B_n$

$\rho_S(R)$ changes:

- only the **relation name** to S

$\rho_{(B_1, \dots, B_n)}(R)$ changes:

- only the **column (attribute) names** to $B_1, \dots, B_n$

15

The UNION Operator $R \cup S$

- $R \cup S$ is a relation that includes all tuples that are either in R or in S (or in both)
- Duplicate tuples are eliminated
- The two operands R and S must be *type compatible*:
 - R and S must have same number of attributes
 - Each pair of corresponding attributes must be type compatible (have same or compatible domains)
- The resulting relation has the same attribute names as the **first** operand R (by convention)

16

UNION

Example:

Retrieve the social security numbers of all employees who either **work in department 5** (RESULT1) or **directly supervise an employee who works in department 5** (RESULT2)

$DEP5EMPS \leftarrow \sigma_{Dno=5}(EMPLOYEE)$

$RESULT1 \leftarrow \pi_{Ssn}(DEP5EMPS)$

$RESULT2 \leftarrow \pi_{Super_ssn}(DEP5EMPS)$

$RESULT \leftarrow RESULT1 \cup RESULT2$

DEP5EMPS

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5

RESULT1

Ssn
123456789
333445555
666884444
453453453

RESULT2

Super_ssn
333445555
888665555

RESULT

Ssn
123456789
333445555
666884444
453453453
888665555

17

The INTERSECTION Operator $R \cap S$

- $R \cap S$ is a relation that includes all tuples that are both in R and in S
- The two operands R and S must be *type compatible*:
 - R and S must have same number of attributes
 - Each pair of corresponding attributes must be type compatible (have same or compatible domains)
- The resulting relation has the same attribute names as the **first** operand R (by convention)

18

The MINUS Operator $R - S$

- $R - S$ is a relation that includes all tuples that are in R but **not** in S
- The two operands R and S must be *type compatible*:
 - R and S must have same number of attributes
 - Each pair of corresponding attributes must be type compatible (have same or compatible domains)
- The resulting relation has the same attribute names as the **first** operand R (by convention)

19

Example

STUDENT and INSTRUCTOR are compatible relations:

STUDENT

F _n	L _n
Susan	Yao
Ramesh	Shah
Johnny	Kohler
Barbara	Jones
Amy	Ford
Jimmy	Wang
Ernest	Gilbert

INSTRUCTOR

F _{name}	L _{name}
John	Smith
Ricardo	Browne
Susan	Yao
Francis	Johnson
Ramesh	Shah

STUDENT $\cup$
INSTRUCTOR

F _n	L _n
Susan	Yao
Ramesh	Shah
Johnny	Kohler
Barbara	Jones
Amy	Ford
Jimmy	Wang
Ernest	Gilbert
John	Smith
Ricardo	Browne
Francis	Johnson

STUDENT $\cap$
INSTRUCTOR

F _n	L _n
Susan	Yao
Ramesh	Shah

STUDENT $-$
INSTRUCTOR

F _n	L _n
Johnny	Kohler
Barbara	Jones
Amy	Ford
Jimmy	Wang
Ernest	Gilbert

20

Properties of UNION, INTERSECT, and MINUS

Commutativity:

$$R \cup S = S \cup R$$
$$R \cap S = S \cap R$$

$$R - S \neq S - R$$

Associativity:

$$(R \cup S) \cup T = R \cup (S \cup T)$$
$$(R \cap S) \cap T = R \cap (S \cap T)$$

21

The CARTESIAN (CROSS PRODUCT) Operator $R \times S$

Combines tuples from two relations in a combinatorial fashion:

$$R(A_1, \dots, A_n) \times S(B_1, \dots, B_m)$$

Result is a relation Q with $n + m$ attributes:

$$Q(A_1, \dots, A_n, B_1, \dots, B_m)$$

in that order.

- Q has one tuple for each *combination* of tuples - one from R and one from S .
- If R has n_R tuples and S has n_S tuples, then $R \times S$ will have $n_R \cdot n_S$ tuples.
- The two operands do NOT have to be *type compatible*.

22

CROSS PRODUCT

Example: List all female employees and their dependents.

$FEMALE_EMPS \leftarrow \sigma_{Sex='F'}(EMPLOYEE)$

$EMP_NAMES \leftarrow \pi_{Fname,Lname,Ssn}(FEMALE_EMPS)$

$EMP_DEPENDENTS \leftarrow EMP_NAMES \times DEPENDENT$

$ACTUAL_DEPS \leftarrow \sigma_{Ssn=Essn}(EMP_DEPENDENTS)$

$RESULT \leftarrow \pi_{Fname,Lname,Dependent_name}(ACTUAL_DEPS)$

All pairs

23

$FEMALE_EMPS \leftarrow \sigma_{Sex='F'}(EMPLOYEE)$

FEMALE_EMPS

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
Alicia	J	Zelaya	999887777	1968-07-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5

$EMP_NAMES \leftarrow \pi_{Fname,Lname,Ssn}(FEMALE_EMPS)$

EMP_NAMES

Fname	Lname	Ssn
Alicia	Zelaya	999887777
Jennifer	Wallace	987654321
Joyce	English	453453453

24

$EMP_DEPENDENTS \leftarrow EMPNAMES \times DEPENDENT$

EMPNAMES

Fname	Lname	Ssn
Alicia	Zelaya	999887777
Jennifer	Wallace	987654321
Joyce	English	453453453

DEPENDENT

Essn	Dependent_name	Sex	Bdate	Relationship
333445555	Alice	F	1986-04-05	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

EMP_DEPENDENTS

Fname	Lname	Ssn	Essn	Dependent_name	Sex	Bdate	...
Alicia	Zelaya	999887777	333445555	Alice	F	1986-04-05	...
Alicia	Zelaya	999887777	333445555	Theodore	M	1983-10-25	...
Alicia	Zelaya	999887777	333445555	Joy	F	1958-05-03	...
Alicia	Zelaya	999887777	987654321	Abner	M	1942-02-28	...
Alicia	Zelaya	999887777	123456789	Michael	M	1988-01-04	...
Alicia	Zelaya	999887777	123456789	Alice	F	1988-12-30	...
Alicia	Zelaya	999887777	123456789	Elizabeth	F	1967-05-05	...
Jennifer	Wallace	987654321	333445555	Alice	F	1986-04-05	...
Jennifer	Wallace	987654321	333445555	Theodore	M	1983-10-25	...
Jennifer	Wallace	987654321	333445555	Joy	F	1958-05-03	...
Jennifer	Wallace	987654321	987654321	Abner	M	1942-02-28	...
Jennifer	Wallace	987654321	123456789	Michael	M	1988-01-04	...
Jennifer	Wallace	987654321	123456789	Alice	F	1988-12-30	...
Jennifer	Wallace	987654321	123456789	Elizabeth	F	1967-05-05	...
Joyce	English	453453453	333445555	Alice	F	1986-04-05	...
Joyce	English	453453453	333445555	Theodore	M	1983-10-25	...
Joyce	English	453453453	333445555	Joy	F	1958-05-03	...
Joyce	English	453453453	987654321	Abner	M	1942-02-28	...
Joyce	English	453453453	123456789	Michael	M	1988-01-04	...
Joyce	English	453453453	123456789	Alice	F	1988-12-30	...
Joyce	English	453453453	123456789	Elizabeth	F	1967-05-05	...

25

$ACTUAL_DEPS \leftarrow \sigma_{Ssn=Essn}(EMP_DEPENDENTS)$

ACTUAL_DEPENDENTS

Fname	Lname	Ssn	Essn	Dependent_name	Sex	Bdate	...
Jennifer	Wallace	987654321	987654321	Abner	M	1942-02-28	...

$RESULT \leftarrow \pi_{Fname,Lname,Dependent_name}(ACTUAL_DEPS)$

RESULT

Fname	Lname	Dependent_name
Jennifer	Wallace	Abner

26

The JOIN Operator $R \bowtie_{cond} S$

- The sequence of CROSS PRODUCT followed by SELECT is used to identify and select related tuples from two relations
- The JOIN operation combines this sequence into a single operation
- JOIN allows to **combine related tuples** from different relations

The JOIN operation on two relations
 $R(A_1, \dots, A_n)$ and $S(B_1, \dots, B_m)$ is:

$$R \bowtie_{cond} S \equiv \sigma_{cond}(R \times S)$$

27

JOIN

Example: List the names of the department managers.

To get the manager's name, we need to combine each DEPARTMENT tuple with the EMPLOYEE tuple whose Ssn value matches the Mgr_ssn value in the DEPARTMENT tuple:

$$\text{DEPT_MGR} \leftarrow \text{DEPARTMENT} \bowtie_{\text{Mgr_ssn}=\text{Ssn}} \text{EMPLOYEE}$$

DEPT_MGR

Dname	Dnumber	Mgr_ssn	...	Fname	Minit	Lname	Ssn	...
Research	5	333445555	...	Franklin	T	Wong	333445555	...
Administration	4	987654321	...	Jennifer	S	Wallace	987654321	...
Headquarters	1	888665555	...	James	E	Borg	888665555	...

28

Properties of JOIN

$$R(A_1, \dots, A_n) \bowtie_{R.A_i=S.B_j} S(B_1, \dots, B_m)$$

The result is a relation Q with $n + m$ attributes:

$$Q(A_1, \dots, A_n, B_1, \dots, B_m) \text{ in that order.}$$

Q has one tuple for each combination of tuples from R and S that **satisfy the join condition** $R[A_i] = S[B_j]$.

If R has n_R tuples, and S has n_S tuples, then $R \bowtie S$ will generally have **less than** $n_R \cdot n_S$ tuples, as only tuples satisfying the join condition will appear in the result:

$$R(A_1, \dots, A_n) \bowtie_{R.A_i=S.B_j} S(B_1, \dots, B_m) = \sigma_{R.A_i=S.B_j} (R(A_1, \dots, A_n) \times S(B_1, \dots, B_m))$$

29

EQUIJOIN

The most common use of join involves conditions with **equality comparisons** only

In the result of an EQUIJOIN there always are one or more pairs of attributes (whose names need not be identical) that have identical values in every tuple.

$$R(A_1, \dots, A_n) \bowtie_{R.A_i=S.B_j} S(B_1, \dots, B_m)$$

30

NATURAL JOIN $R * S$

NATURAL JOIN eliminates the second (superfluous) attribute in an EQUIJOIN.

NATURAL JOIN requires that each pair of corresponding join attributes **have the same name** in both relations.

31

NATURAL JOIN

Example:

$DEPT_LOCS \leftarrow DEPARTMENT * DEPT_LOCATIONS$

Since the only attribute with the same name is Dnumber, an implicit join condition is created based on this attribute:

$DEPARTMENT.Dnumber = DEPT_LOCATIONS.Dnumber$

DEPT_LOCS

Dname	Dnumber	Mgr_ssn	Mgr_start_date	Location
Headquarters	1	888665555	1981-06-19	Houston
Administration	4	987654321	1995-01-01	Stafford
Research	5	333445555	1988-05-22	Bellaire
Research	5	333445555	1988-05-22	Sugarland
Research	5	333445555	1988-05-22	Houston

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

DEPT_LOCATIONS

Dnumber	Dlocation
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

32

NATURAL JOIN

Example:

$$Q \leftarrow R(A, B, C, D) * S(C, D, E)$$

The implicit join condition includes the AND of equating each pair of attributes with the same name:

$$R.C = S.C \text{ AND } R.D = S.D$$

The result keeps only one attribute of each such pair:

$$Q(A, B, C, D, E)$$

33

Complete Set of Relational Operators

The set of operations SELECT, PROJECT, UNION, DIFFERENCE, RENAME, and CROSS PRODUCT is a **complete set**:

Any other relational algebra expression can be expressed by a combination of these five operators.

Examples:

$$R \cap S = (R \cup S) - ((R - S) \cup (S - R))$$

$$R \bowtie_{cond} S = \sigma_{cond}(R \times S)$$

34

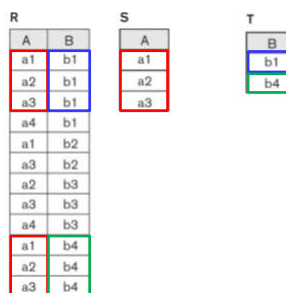
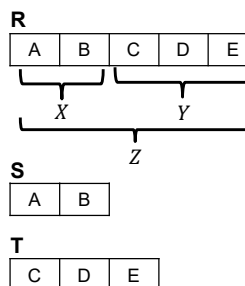
The DIVISION Operator

$T(Y) = R(Z) \div S(X)$ where $X \subseteq Z$.

An attempt to find T such that $R = S \times T$

Why not take $T = \pi_Y(R)$?

But then usually $R \supset S \times T$, so we make do with the maximal T such that $R \subseteq S \times T$



35

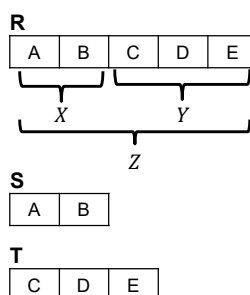
The DIVISION Operator

$T(Y) = R(Z) \div S(X)$ where $X \subseteq Z$.

Let $Y = Z - X$ (i.e. Y is the set of attributes of R that are not attributes of S).

$T(Y)$ includes a tuple t if tuples t_R appear in R with $t_R[Y] = t$ and $t_R[X] = t_S$ for **every** tuple $t_S \in S$.

$R(Z) \subseteq S(X) \times T(Y)$



36

DIVISION

Implements a “for all” type of query:

“Find the rows in R that correspond to **all** rows of S ”

$$T \leftarrow R \div S$$

R		S		T	
A	B	A		B	
a1	b1	a1		b1	
a2	b1	a2		b4	
a3	b1	a3			
a4	b1				
a1	b2				
a3	b2				
a2	b3				
a3	b3				
a4	b3				
a1	b4				
a2	b4				
a3	b4				

37

DIVISION

Example: List the employees who work on **all** the projects that John Smith works on.

Step 1: List all the project numbers that John Smith works on.

$SMITH \leftarrow \sigma_{Fname='John' \text{ AND } Lname='Smith'}(EMPLOYEE)$

$SMITH_PNOS \leftarrow \pi_{Pno}(WORKS_ON \bowtie_{Essn=SSn} SMITH)$

SMITH_PNOS	
Pno	
1	
2	

38

DIVISION

SMITH_PNOS
Pno
1
2

SSN_PNOS
Essn Pno
123456789 1
123456789 2
666884444 3
453453453 1
453453453 2
333445555 2
333445555 3
333445555 10
333445555 20
999887777 30
999887777 10
987987987 10
987987987 30
987654321 30
987654321 20
888665555 20

Step 2: Trim down WORKS_ON

$SSN_PNOS \leftarrow \pi_{Essn, Pno}(WORKS_ON)$

Step 3: List the SSN's of the employees who work on ALL these projects

$SSNS(SSn) \leftarrow SSN_PNOS \div SMITH_PNOS$

rename

SSNS
Ssn
123456789
453453453

Step 4: List the names of the employees with these SSN's

$RESULT \leftarrow \pi_{Fname, Lname}(SSNS * EMPLOYEE)$

Fname	Lname
John	Smith
Joyce	English

39

DIVISION

The DIVISION operator can be expressed using the π , $\times$, and $-$ operators:

$$T1 \leftarrow \pi_B(R)$$

$$T2 \leftarrow \pi_B((S \times T1) - R)$$

$$T \leftarrow T1 - T2$$

So

$$R \div S = \pi_B(R) - \pi_B((S \times (\pi_B(R))) - R)$$

R
A B
a1 b1
a2 b1
a3 b1
a4 b1
a1 b2
a3 b2
a2 b3
a3 b3
a4 b3
a1 b4
a2 b4
a3 b4

S
A
a1
a2
a3

T
B
b1
b4

SxT1
A B
a1 b1
a2 b1
a3 b1
a1 b2
a2 b2
a3 b2
a1 b3
a2 b3
a3 b3
a1 b4
a2 b4
a3 b4

T1
B
b1
b2
b3
b4

(SxT1)-R
A B
a2 b2
a1 b3

T2
B
b2
b3

40

DIVISION in SQL

Query: List first and last names of employees who work on **ALL** projects controlled by department #4.

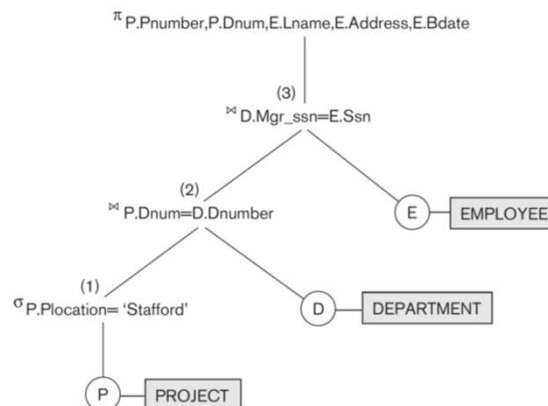
```
SELECT E.Fname, E.Lname
FROM   EMPLOYEE E
WHERE  NOT EXISTS (SELECT Pnumber
                   FROM   PROJECT P
                   WHERE  P.Dnum = 4
                   EXCEPT
                   SELECT Pno
                   FROM   WORKS_ON W
                   WHERE  W.Essn=E.Ssn);
```

Lname	Fname
Alicia	Zalaya
Ahmad	Jabbar

41

Query Tree

```
SELECT P.Pnumber, P.Dnum, E.Lname,
       E.Address, E.Bdate
FROM   PROJECT P, DEPARTMENT D, EMPLOYEE E,
WHERE  P.Plocation = 'Stafford' AND
       P.Dnum = D.Dnumber AND
       D.Mgr_ssn = E.Ssn
```



42

LEFT OUTER JOIN

Example: List **all** the employees with the names of the departments they manage.
Use NULL if they do not manage a department.

$\pi_{Fname, Lname, Dname} (EMPLOYEE \bowtie_{Ssn=M_ssn} DEPARTMENT)$

Fname	Lname	Dname
John	Smith	NULL
Franklin	Wong	Research
Alicia	Zelaya	NULL
Jennifer	Wallace	Administration
Ramesh	Narayan	NULL
Joyce	English	NULL
Ahmad	Jabbar	NULL
James	Borg	Headquarters

43

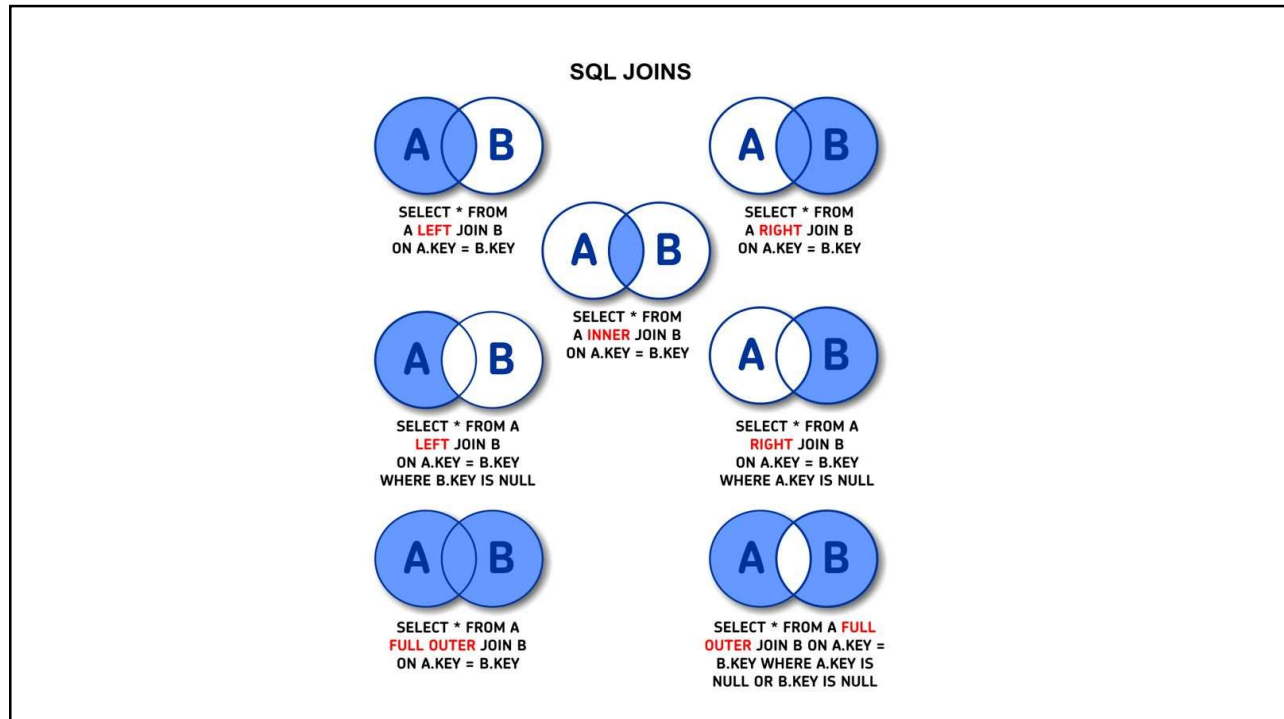
OUTER JOIN

LEFT OUTER JOIN: $R \bowtie S$ keeps **every tuple** in the first relation R . If no matching tuple is found in S , the attributes of S in the result are filled or “padded” with null values.

RIGHT OUTER JOIN: $R \bowtie S$ keeps **every tuple** in the second relation S . If no matching tuple is found in R , the attributes of R in the result are filled or “padded” with null values.

FULL OUTER JOIN: $R \bowtie S$ keeps **every tuple** in **both** relations. If no matching tuple is found in one relation, the corresponding attributes of the other in the result are filled or “padded” with null values.

44



45

Aggregate Functions

Functions applied to collections of numeric values:

- COUNT
- SUM
- AVERAGE
- MAXIMUM
- MINIMUM

Syntax: $\mathcal{F}_{\langle function_list \rangle} (R)$

Example: $\mathcal{F}_{\text{COUNT } Ssn, \text{AVERAGE } Salary}(\text{EMPLOYEE})$

Note: COUNT counts the number of rows, without removing duplicates

46

Aggregate Functions With Grouping

Example: For each department, retrieve the Dno, COUNT Ssn, and AVERAGE Salary

$\rho_{R(Dno, Nemp, Avsal)}(Dno \mathcal{F}_{COUNT Ssn, AVERAGE Salary}(EMPLOYEE))$

Fname	Minit	Lname	Ssn	...	Salary	Super_ssn	Dno
John	B	Smith	123456789		30000	333445555	5
Franklin	T	Wong	333445555		40000	888665555	5
Ramesh	K	Narayan	666884444		38000	333445555	5
Joyce	A	English	453453453	...	25000	333445555	5
Alicia	J	Zelaya	999887777		25000	987654321	4
Jennifer	S	Wallace	987654321		43000	888665555	4
Ahmad	V	Jabbar	987987987		25000	987654321	4
James	E	Bong	888665555		55000	NULL	1

Grouping EMPLOYEE tuples by the value of Dno

Dno	Count (*)	Avg (Salary)
5	4	33250
4	3	31000
1	1	55000

Dno	Nemp	Avsal
5	4	33250
4	3	31000
1	1	55000

47

Example

Query: Retrieve the names and addresses of all employees who work for the 'Research' department.

$RDEPT \leftarrow \sigma_{Dname = 'Research'}(DEPARTMENT)$

$REMPS \leftarrow (RDEPT \bowtie_{Dnumber=Dno} EMPLOYEE)$

$RESULT \leftarrow \pi_{Fname, Lname, Address}(REMPS)$

As one query:

$\pi_{Fname, Lname, Address}(\sigma_{Dname = 'Research'}(DEPARTMENT) \bowtie_{Dnumber=Dno} EMPLOYEE)$

Fname	Lname	Address
John	Smith	731 Fondren, Houston, TX
Franklin	Wong	638 Voss, Houston, TX
Ramesh	Narayan	975 Fire Oak, Humble, TX
Joyce	English	5631 Rice, Houston, TX

48

Example

Query: Retrieve the names of employees who have no dependents.

$AEMP \leftarrow \pi_{Ssn}(EMPLOYEE)$

$EWD(Ssn) \leftarrow \pi_{Essn}(DEPENDENT)$

$EWOD \leftarrow AEMP - EWD$

$RESULT \leftarrow \pi_{Fname,Lname}(EWOD * EMPLOYEE)$

Fname	Lname
Alicia	Zalaya
Narayan	Ramesh
Joyce	English
James	Borg

As one query:

$\pi_{Fname,Lname} \left(\left(\pi_{Ssn}(EMPLOYEE) - \pi_{Essn}(DEPENDENT) \right) * EMPLOYEE \right)$